Candidate Name: ____________________

Class	Adm No

H

millennia institute

2018 Preliminary Exams
Pre-University 3

BIOLOGY **9744/03**

Paper 3 Long Structured and Free-response Questions **17 September 2018**

2 hours

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Admission number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer all questions in the space provided on the Question Paper.

Section B
Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
Section B	
Total	

This question paper consists of 22 printed pages, including 1 blank page.

[Turn over

Section A

Answer **all** questions in this section.

1. Monoclonal antibodies are antibodies derived from a single unique parent cell. In the industry, monoclonal antibodies are produced from hybridoma. The procedures are illustrated in Fig. 1.1.

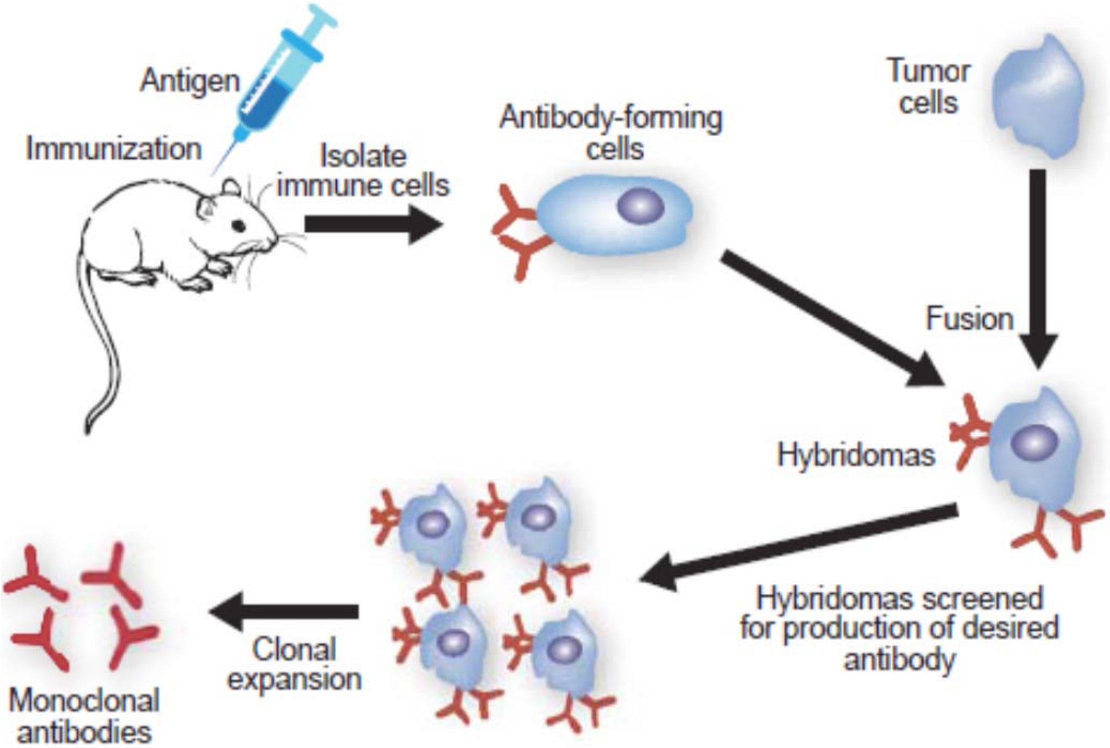


Fig. 1.1

The steps to produce monoclonal antibodies using hybridoma technique are as follow:

Step 1: A mouse is injected with a specific antigen.
Step 2: The mouse is left for a few weeks to allow immune response to occur.
Step 3: Immune cells are isolated and plasma cells are extracted from the mouse's spleen.
Step 4: Plasma cells are fused with tumour cells to form hybridoma.
Step 5: Hybridomas are screened to select the ones that are producing the desired antibody.
Step 6: Selected hybridomas are isolated to grow and divide during clonal expansion.

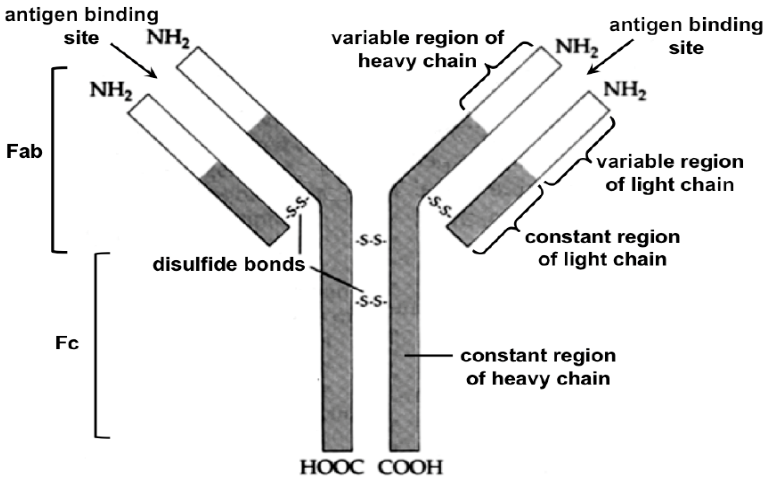
(a) State the type of immunity involved in the production of monoclonal antibody.

- Active artificial immunity

…………………………………………………………………………………………………**[1]**

(b) In the space below, draw a labelled diagram of an antibody such as immunoglobulin G (IgG).

- Correct shape;
- Correctly label and number of light and heavy chains;
- Correctly label and number of variable and constant regions;
- Fab and Fc;



[3]

(c) Describe the events that occur in step 2 during the immune response in the hybridoma technique.

- Antigen is recognised as non-self/foreign by B cell/lymphocyte/APC;
- B cell with specific antigen receptor complementary to the antigen;
- B cell phagocytose the pathogen with the antigen;
- And displays class II MHC:peptide complex on the cell surface membrane of B cell;
- Activated CD4+ T helper cells with the same specific receptor will bind to the complex;
- And activate B cells which will then undergo clonal expansion to form more plasma and memory B cells/lymphocytes;

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Hybridoma is a hybrid cell consisting of both tumour and plasma cells.

(d) Explain how mutations result in cancer.

- Gain-of-function mutation of proto-oncogene gene result in mutated oncogene;
  OR
- Gain-of-function mutation of ras gene result in mutated hyperactive/OWTTE Ras protein;
  OR
- Gain-of-function mutation of *c-myc* gene result in mutated hyperactive/OWTTE c-myc protein;
- Gain-of-function mutation is dominant where only one mutated allele is needed;
- Leads to over-stimulation of cell cycle through constitutive active protein/protein binding to substrate with increased affinity/protein being resistant to degradation;

Max 2

- Loss-of-function mutation of tumour suppressive gene;
  OR
- Loss-of-function mutation of p53 gene result in mutated inactivated p53 protein;
- Loss-of-function mutation is recessive where two copies of the alleles need to be mutated to produce non-functional tumour suppressive protein;
- Leads to loss of cell cycle arrest through loss of ability for DNA repair/loss of apoptosis during DNA damage/loss of cell cycle control/inappropriate cell cycle progression;

- Accumulation of mutations results in uncontrolled cell division;

Max 2

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4]

(e) Identify two such factors.

- Carcinogenic chemicals such as tar, ethidium bromide and tetrachloromethane
- Ionizing radidation such as ultraviolet radiation from sunlight
- Loss of immunity/immunosuppression that may weakens the immune system, thus increased chance of developing cancer
- Integration of viral DNA and disrupting proto-oncogene/tumor suppressor gene
- Genetic disposition/inherited in the family
  Two factors to score one mark;

....

...........[1]

The tumour cells were genetically modified such that an enzyme known as hypoxanthine-guanine-phosphoribosyltransferase (HGPRT) is absent. HGPRT enzyme is involved in the synthesis of purine. On the other hand, the plasma cells have the HGPRT enzyme. Therefore, the fused plasma cell in the hybridoma supplies the HGPRT enzyme for the survival of the hybridomas.

(f) Explain the roles of purine in living organisms.

- Purine is a type of nitrogenous base that form part of the structure of nucleotide;
- Adenine and guanine are examples of purines;
- Forms deoxyadenosine 5'-monophosphate and deoxyguanosine 5-monophosphate in DNA;
- Forms riboadenosine 5-monophosphate and riboguanosine 5-monophosphate in RNA;
- Form the energy currency ATP and GTP;
- Important biomolecules/cofactor/reducing agent such as NADPH / NADH;

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.**[2]**

(g) Explain why unfused tumour cells and unfused plasma cells are unable to grow for a long period of time.

- Unfused tumor cells do not have the HGPRT enzyme to synthsise purine, which is essential for metabolism/survival/DNA replication/gene expression of cell/AVP;
- Unfused plasma cells are unable to grow indefinitely due to end-replication problem/absence of telomerase;

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....**[2]**

The HGPRT enzyme consists of four polypeptide chains, as shown in Fig. 1.2.

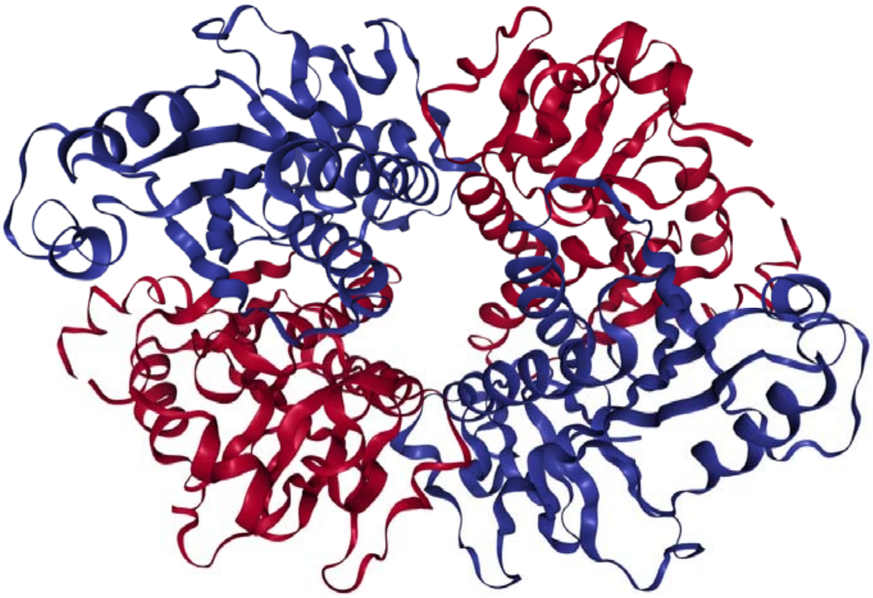

Fig. 1.2

(h) Explain the importance of the shape of HGPRT enzyme to its function.

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……[2]

- Globular;

Any 1 of the following

- All HGPRT enzymes/proteins expressed have the same amino acid sequence and thus have the same 3-dimensional conformation, giving the same binding site;
- Soluble in water and thus able to perform its function in the cell's aqueous environment;
- Amino acid with polar and charged R groups are found on the exterior surface of protein/enzyme;
- Amino acid with hydrophobic R groups are found on the interior surface of protein/enzyme;
- AVP;

Rheumatoid arthritis (RA) is an autoimmune disease in which T lymphocytes attack the cartilage of joints by secreting a protein, TNFα. RA can be treated using either the anti-inflammatory drug, MTX, or monoclonal antibody, infliximab.

Five groups of people with RA received the following treatments for one year:

- group **P** – MTX only
- group **Q** – MTX plus low dosage of infliximab at intervals of eight weeks
- group **R** – MTX plus low dosage of infliximab at intervals of four weeks
- group **S** – MTX plus high dosage of infliximab at intervals of eight weeks
- group **T** – MTX plus high dosage of infliximab at intervals of four weeks.

At the end of the year's treatment, the proportion of people in each group with increased joint damage was determined.

The experimental results are shown in Fig. 1.3. The number of people in each group is shown in brackets. All bar graphs display means $\pm$ standard error mean (S.E.M) (sample size $\geqslant 3$). Comparisons between infliximab treatments and untreated controls were made using a Student's t-test (*p < 0.5, **p < 0.01, ***p < 0.001).

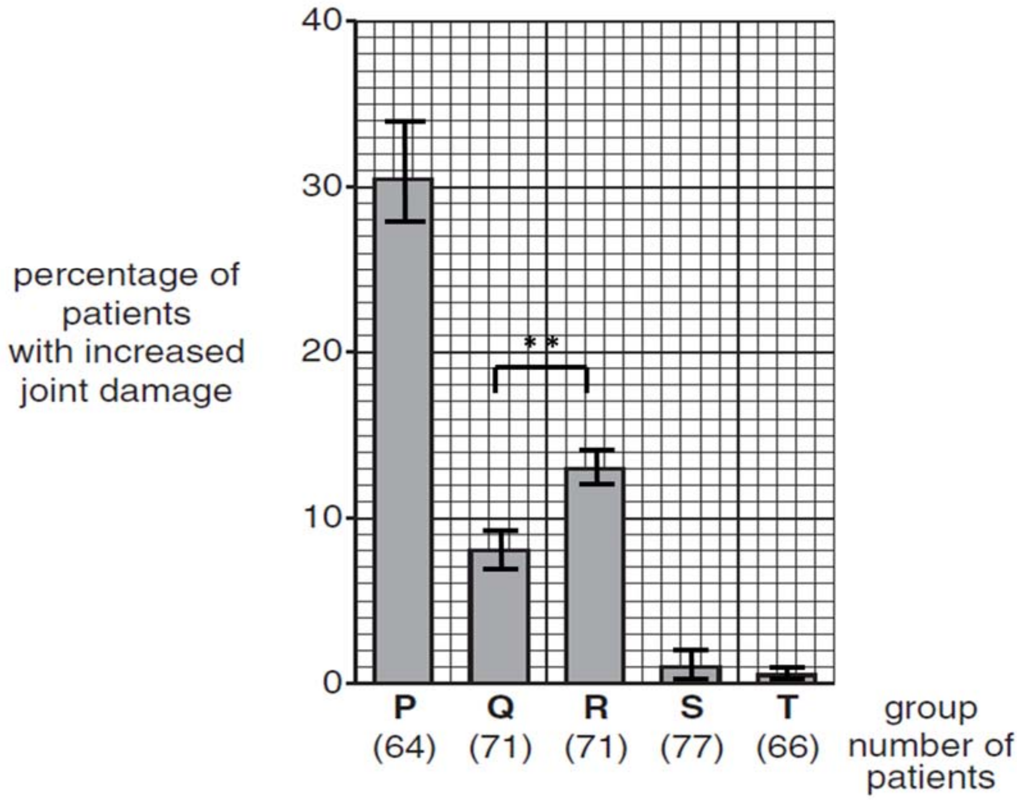

Fig. 1.3

(i) Discuss the effectiveness of infliximab treatment on these patients.

- in group P patients without infliximab treatment serving as control, the percentage of patients with increased join damage is highest at 30%;
- The percentages of patients with increased joint damage after given low dosage of infliximab are 8% and 13% for doses given at eight weeks interval and four weeks intervals respectively;
- The percentages of patients with increased joint damage after given high dosage of infliximab are 1% and 0.5% for doses given at eight weeks interval and four weeks intervals respectively;

Max 2

- There was higher percentage of patients with increased joint damage in patients given low dosage of infliximab than patients given high dosage of infliximab;
  OR
- At high dosage of infliximab treatment, there are lesser patients with increased joint damage as compared to low dosage of infliximab treatment;

- In the presence of infliximab, there are lesser patients with increased joint damage;
- High dosage given at either four or eight weeks interval is (most) effective;

…… …… …… …… …… …… …… …… ……
..**[4]**

(j) With reference to the S.E.M and t-test results, assess the validity of the experimental results.

- There is no significant difference between group S and group T;
- since the standard error bar of S and T overlap;
- Imply that the effect of high dosage is the same at intervals of eight weeks and at four weeks;
- Group Q should have higher percentage of patients with increased joint damage since it is administered at a lower frequency than R;
- There is significant difference between group Q and group R;
- since the standard error bar of S and T overlap;
- AVP;

…….. …….. ……..
….**[2]**

al:25]

2. The middle of the Nevada desert seems an unlikely spot to find a fish. But an underground fissure in scorching-hot Death Valley is the only natural habitat for the endangered Devils Hole pupfish, a silvery blue creature about the size of a pet goldfish.

The water inside Devils Hole is consistently a toasty 32°C, hot enough to kill most other fish. For 2 months of the year, the cavern—which opens into a deep subterranean lake—receives no direct sunlight. And yet scientists believe a small population of pupfish has lived there for 10,000 to 20,000 years.

Now, evidence is growing that these fish might be far younger than previously assumed: A new study suggests that the Devils Hole pupfish actually colonised its watery cavern somewhere between 105 and 830 years ago, making scientists rethink how it got there in the first place. "Our new estimate of the younger age in a way makes this species even more fascinating because it has so many of these unique traits relative to other pupfish," says Christopher Martin, a biologist at the University of North Carolina, Chapel Hill, who led the study.

Pupfish first arrived in Death Valley during a wet period, probably when water joined the area to lakes or rivers elsewhere. But as the water dried up and the area turned to desert, the pupfish became isolated in a smattering of springs.

Devils Hole is especially isolated, and the pupfish population there is particularly small, ranging from 35 to 548 fish in the remote cavern since official recordkeeping began in the 1970s. Their survival has been heralded as an evolutionary anomaly, as such populations usually become severely inbred and die out. "All we know about conservation genetics suggests that these populations of very small size should not be able to survive in the long term," Martin says.

To figure out just when the Devils Hole pupfish diverged from its kin, the researchers sequenced 13,000 different stretches of DNA from 56 pupfish from around Death Valley and the world. Those data allowed them to reconstruct the area's pupfish family tree and calculate when the different species emerged. Although scientists previously believed the first pupfish species came to Death Valley several million years ago, these analyses suggest they arrived around the time of the valley's most recent flood, just 10,000 years ago. The analysis also suggests the Devils Hole pupfish became isolated from other pupfish in Death Valley fewer than a thousand years ago, much more recently than expected.

http://www.sciencemag.org/news/2016/01/bizarre-desert-dwelling-fish-may-have-evolved-just-couple-hundred-years-ago. Extracted on 10th June 2018.

(a) Suggest why there is discrepancy to the age of the early ancestor of pupfish.

- Tentative nature of knowledge due to lack of knowledge in the past ;
- Using recent technology deepens knowledge;
- AVP;

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…..**[1]**

Scientists believed that the change from an extensive lake system to a few pools drives the evolution of five species of the pupfish.

(b) Explain how this may happen.

- Existing <u>variation</u> exists within the <u>ancestral/OWTTE population</u> due to <u>random mutations;</u>
- <u>Different selection pressures</u> in each area where each population resides;
- Fish with <u>alleles</u> that confer <u>selective advantage</u> survive to reproductive age, and <u>pass on these alleles</u> to offspring (viable and fertile);
- Results in <u>change in allele frequency / gene pool,</u> leading to <u>allopatric speciation</u>
- <u>Reproductive isolation</u> resulting in <u>absence of gene flow;</u>
- The isolated population begins to evolve independently (idea) via genetic drift, mutation);
- Accumulation of sufficient genetic differences/mutation, hence <u>unable to interbreed to produce viable and fertile offspring;</u>
- <u>Thus, altering the allele frequencies in each isolated population;</u>

…………………………………………………………………………………**[5]**

During autumn in 2017, there were only 115 of the fishes in the Death Valley.

(c) Suggest why the pupfish becomes endangered when their population falls to very low numbers.

- There will be difficulty in finding a <u>mate</u> and <u>reproduce</u>;
- <u>Smaller gene pool/loss in genetic diversity</u> and thus more vulnerable to <u>changes</u> in environment;
- The rarity of species may attract poachers;
- Difficult/slower for population number to increase significantly;
- AVP;

……..**[2]**

In the study, the scientists used biochemical data to establish the evolutionary relationship of the pupfish.

(d) Describe two other methods that the scientists could use to understand the homology between different fish species.

- Embryological homology comparing structure of early development;
- Fossil record of fishes preserved in sedimentary rocks;
- Biogeography to study the geographical distribution of species;
- Anatomical homology to study homologous or analogous body structures between species;

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The pupfish survives in water with high temperature of 32^{o}C, minimal food, and receive no sunlight two months of the year.

(e) Describe the effect of temperature on protein structure and function.

- Heat increases the kinetic energy of the molecules, breaking weak bonds;
- such as ionic bond, hydrogen bond and hydrophobic interaction(any two) that maintain the specific 3-dimesional shape of the protein;
- resulting in denaturation and loss of protein function;
- Aggregation of unfolded protein may take place;

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(f) Explain how the biomolecules of organisms adapt to the harsh living condition of Death Valley.

- Proteins contain strong disulphide bonds between the;
- R groups of amino acids holding the tertiary and quaternary structure;
- Carbohydrates such as cellulose contain many cross linkages;
- Triglycerides contain fatty acid with saturated carbon-carbon bond;
- Nucleic acids contain more G=C hydrogen bonds than A=T hydrogen bonds;
- AVP;

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……………………………………………………………………………………………[3]

[Turn over

The Death Valley receives less than two inches of rain every year. In comparison, the average rainfall in the United States is 32 inches in 2017.

(g) Predict what may happen to the pupfish if water level rises to form an extensive lake system.

- Number of fishes of all species may increase initially;
- Due to increased breeding space/food;
- Eventually, there may be competition between the five species of pupfish;
- Possibly reducing the number of fishes in long term;
- The different species of fishes experience different selection pressure;
- Thus, not all fishes may survive/ORA;
- Interbreeding between species with compatible genetics/chromosome number;
- Different species may occupy different niches within the lake system;

[2]

Rising sea level may be caused by the addition of extra water from ice sheet or the thermal expansion of sea water. Thermal expansion of sea water refers to the increase in volume of sea water due to the increased sea temperature.

(h) Describe three other effects of increased sea temperature.

- Increase in temperature in the ocean water may stress the coral and zooxanthellae symbiotic relationship, leading to coral bleaching;
- Increased evaporation may lead to increased precipitation/heavy rain;
- Leads to rising sea level may results in flooding;
- Flooding may stress the fresh water supplies;
- Migration of marine life where temperature is cooler and more optimal for their survival/ORA;

[3]

Besides the increase in sea temperature, another effect of global warming is the increase in surface temperature. Fig. 2.1 shows the annual mean surface temperature in Singapore from 1948 to 2011.

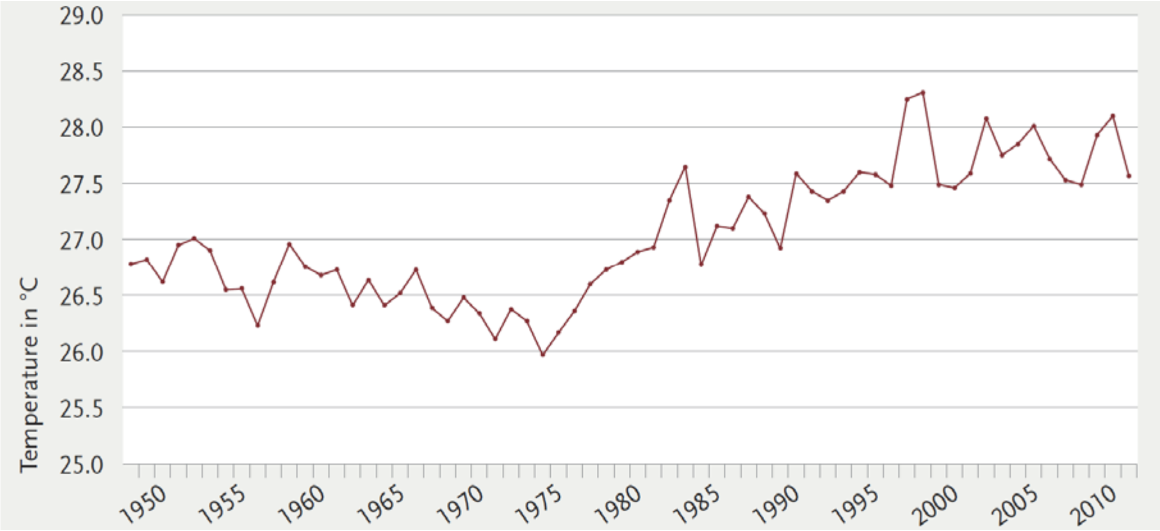

Fig. 2.1

(i) Describe the changes in annual mean surface temperature from 1948 to 2011.

- The annual mean surface temperature decreases in fluctuating manner from 26.8°C in 1948 to 26.0 °C in 1975/decrease by 8 °C from 1948 to 1975;
- The annual mean surface temperature increases from 26.0 °C in 1975 to 26.9 °C in 1981/1982;
- Further increase in fluctuating manner to 27.6 °C in 2011;

....**[3]**

(j) Suggest the event that may take place in 1975.

- Urbanisation/increase number of building;
- Increase emission of greenhouse gases/carbon dioxide/methane;
- AVP;

....**[1]**

[Total:25]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous pose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

3 **(a)** With reference to a named virus and its reproductive cycle, describe how the virus can transfer genes between bacteria. **[10]**

(b) 'Without genetic variations, antibody is completely useless in the immune system.'

Discuss on the validity of this statement. **[15]**

[Total: 25]

4 **(a)** Plants and other photosynthetic organisms convert light energy into chemical energy in the form of ATP and NADPH. Photosynthesis comprises both light and dark reactions, which are catalysed and regulated by proteins.

Describe the importance of proteins in photosynthesis. **[10]**

(b) Discuss the extent to which increased temperature lead to the increase in dengue transmission. **[15]**

[Total: 25]

3a) With reference to a named virus and its reproductive cycle, describe how the virus can transfer genes between bacteria. **[10]**

1 Lambda phage can undergo specialized transduction;
2 when it switches from a lysogenic to a lytic cycle;.
3 The single tail fibres of the Lambda phage attach to specific receptor sites on the surface of a bacterial host cell;
4 The tail sheath contracts, piercing through the bacterial cell wall and cell membrane and the phage DNA is injected into the bacterial cell;

5 Lambda genome is integrated into specific site on the bacterial chromosome/prophage insertion site via integrase;

6 Once integrated, the phage DNA in a bacterium is called a prophage.
7 A prophage gene could code for a repressor protein that prevents transcription of most of the other prophage genes.

8 When the bacteria cell divides by binary fission, it also replicates the prophage DNA along with its own bacterial chromosome, passing on the prophage to daughter cells;

9 Switch from lysogenic to lytic cycle due to environmental signals, such as radiation, availability of nutrients or the presence of certain chemical (any 2);
10 The repressor proteins that were repressing lambda gene replication are no longer made;
11 allowing the Lambda phage genome to be excised from the bacterial chromosome;
12 The viral genome may not be excised properly at this stage and hence viral DNA may contain bacterial genes which are adjacent to the virus DNA during integration;
13 The virus DNA, containing some of the host bacterial gene, is packed into a new viral particle and released;
14 The viral particle may infect another bacteria and inject its DNA along with the host bacterial DNA;
15 The bacterial DNA may undergo crossing over with the homologous DNA region in the recipient cell, resulting in new combination of alleles/gene in the recipient;

QWC: Good spread of knowledge communicated without ambiguity to include at least 3 marking points from each paragraph

3b) 'Without genetic variations, antibody is completely useless in the immune system.'

Discuss the validity of this statement. **[15]**

Stand [compulsory]

1 Antibody useless to a large extent without genetic recombination;
2 AVP;

Antibody and why antibody is useless

3 Antibody is the soluble form of B cell receptor secreted from activated plasma cells;
4 Genetic variation occurs to generate the wide diversity of antibody;
5 Without these genetic variation, immune system is only useful to a limited amount of antigen and effector function;
6 Foreign antigen will not be recognized as non-self and no immune response can be launched;
7 May lead to uncontrolled infection by pathogen;
8 Moreover, the antigen on pathogen may be mutated, thus it is important to have wide repertoire of antibody;
Max 3

V(D)J recombination and its importance

9 During V(D)J recombination/somatic recombination, the heavy and light chains at the variable region;
10 Undergo gene recombination during the differentiation of B cells;
11 There are multiple copies of the V, D and J gene segments, thus many different variable/V regions can be made by selecting different combinations of these segments;
12 For instance, In the **κ**/kappa light chain, the 40 V_k and 5 J_k gene segments can recombine to generate 200 possible $V_κ$ regions;
13 The choice of light chain and random pairing of the different variable regions of light and heavy chains during formation of the receptors;
14 Results in the highly diverse repertoire/OWTTE antigen binding sites of antibody;

Max 3

Somatic hypermutation and its importance

15 Somatic hypermutation refers to random point mutation;
16 in the rearranged VDJ region of heavy chain gene locus and the rearranged VJ region of light chain locus;
17 of the activated B cells in the adaptive immune response during clonal expansion;
18 Each clone with a mutation acquires an altered amino acid sequence within the variable region of the Ig chains;
19 This leads to affinity maturation, the average increase in affinity of the antibody to the antigen;
Max 3

Class switching and its importance

20 *Class/isotype switching is the process of B cell switching its isotype classes;*

21 regulated by cytokines released by $CD4^+$ T helper cells;

22 *occurs at the constant region of the heavy chain gene locus;*

23 Enables the same VH/variable heavy exon to be associate with different CH/constant heavy exon in the course of an immune response.

24 *Provide antibody with same antigen specificity but distinct/different effector functions;*
Max 2

How antibody can still be useful

25 Without genetic variation, antibody is still able to recognize limited number of epitope through the antibody's antigen binding site;

26 Without genetic variation, the antigen binding site of the antibody can still bind to the limited number of epitope but at lower affinity;

27 The antigen may not be able to bind to the antigen binding site firmly/OWTTE OR The antigen may fall off from the antigen binding site;

28 Without genetic variation, the antibody can still function with the limited type of effector functions;
Max 2

QWC: Good spread of knowledge communicated without ambiguity to include both sides of discussion and reference to the three types of genetic variation;

4a) Plants and other photosynthetic organisms convert light energy into chemical energy in the form of ATP and NADPH. Photosynthesis comprises of both light and dark reactions, catalysed and regulated by proteins.

Describe the importance of proteins in photosynthesis. **[10]**

Light reaction

1 Photosystem is a light–capturing unit located in the thylakoid membrane of the chloroplast;
2 consisting of a reaction centre surrounded by numerous light harvesting complexes.
3 The photosystems are made up of chlorophyll/pigment and protein;
4 The light harvesting complex enables the photosystem to absorb light energy and transfer to the reaction centre;
5 The energy is received by special chlorophyll a in the reaction centre, and thus transferred to the electron acceptor, a protein;
6 During the light dependent reaction, proteins/electron carriers such as cytochrome complex are involved in the electron transport chain;
7 Transfer of electron along protein carriers of ETC releases energy;
8 to pump proton from the stroma to the thylakoid space;
9 the enzyme NADP+ reductase transfers electrons from the protein to NADP+;
10 In chemiosmosis, ATP synthase is a transport protein;
11 Enables the diffusion of hydrogen ions from the thylakoid space into the stroma;
12 This provides energy for the synthesis of ATP from ADP;

Dark reaction

13 The ATP and NADPH are used in the dark reaction/Calvin cycle to synthesis sugar;
14 In the dark rection/calvin cycle, an enzyme Rubisco/rubulose bishosphate catalyse carbon fixation/incorporation of carbon dioxide with ribulose bisphosphate;
15 Many other enzymes are involved in the catalysis of other steps in Calvin cycle;

QWC: Include example and respective roles of proteins involve in light and dark reactions.

Discuss the extent of how increased temperature can lead to the increase in dengue transmission. **[15]**

1. *Aedes aegypti* is the primary vector of dengue virus;
2. Mosquitoes are cold blooded animals where their body temperature is affected by the environment;
3. Embryonic development is usually completed in 48 hours in a warm and humid environment, but may take weeks in cooler environment;
4. Increased temperature leads to shorter hatching time;
5. Larval development occurs in four stages and is dependent on temperature, availability and density;

Max 3

6. Temperature greatly influence its behaviour, development, reproduction and survival;
7. At temperature higher than 30^{o}C, the survival rate of larvae and pupae drop/OWTTE;
8. Increased temperature increases the rate of enzymae catalysed reactions which lead to higher metabolic rates;
9. As metabolic rates increase, the development rates/life cycle of egg, larva and pupa shortens/hasten;
10. At higher temperature, insects will mature, mate and reproduce in a shorter span of time than normal and consequently there is a greater capacity to produce more offspring during the transmission period;
11. At higher temperature, female mosquitoes digest blood faster and feed more frequently, thus increasing transmission intensity;
12. Increased temperature leads to increase water temperature, larvae take shorter time to mature and consequently there is greater capacity to produce more offspring.
13. Shorten developmental time of mosquito increases its survival rate as egg, larvae and pupae are less susceptible to predators, diseases and parasitism.

Max 5

14. Increased temperature encourages mosquitoes to migrate to higher latitudes with colder temperature;
15. Increased temperature may lead to warmer and shorter winters, allowing more mosquitoes to survive during and through winter as they can be active for a longer period of time;
16. In warmer climate, dengue virus complete extrinsic incubation within the female mosquito faster, thereby increasing the proportion of infectious vector;
17. Increased temperature leading to increased precipitation/rainfall increases the number and quality of breeding sites for mosquitoes;

Max 3

How increased temperature does not lead to increased transmission rate

18. At temperature beyond 35^{o}C, the survival rate of mosquito at all stages of life cycle decreases, thus decreasing transmission rate;
19. Due to decrease activity of enzyme at higher temperature, decreasing the rate of enzyme-catalysed reaction;

Max 1

Other factors

20. Economy of country/region will affect the extent of vector control;
21. Drug resistance of dengue virus strain;

22 Dense human population can lead to increase transmission;
23 Migration of people/increasing people movement;
Max 1

Stand [compulsory]
24 Increased temperature can lead to an increase in dengue transmission to a large extent, but can be managed with vector control management;
25 AVP;

QWC: Stand/Conclusion of discussion and coherent link between temperature, life cycle, mosquito and transmission;